An AI-Assisted System for Credit Card Benefit Tracking and Recommendation

RETRIEVAL-AUGMENTED GENERATION FOR PERSONAL FINANCE

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Abstract

Managing credit card benefits presents a significant cognitive challenge: reward structures are complex, offers are time-sensitive, and benefit information is dispersed across issuer portals with no standardized interface. This project presents an AI-assisted system that combines LLM-powered benefit extraction, RAG-based card recommendation, and human-in-the-loop validation to reduce this cognitive burden and support everyday financial decisions. The system integrates structured benefit databases built from unstructured issuer content, screenshot-parsed promotional offers, and real-time geolocation context. Results demonstrate that LLMs provide the greatest value as structured extraction and retrieval tools rather than autonomous decision-makers — with system reliability depending on schema constraints, deterministic validation, and human oversight. The work demonstrates a practical model for grounded, transparent AI in personal finance, with implications for any domain where high-value decisions depend on unstructured, dispersed information.

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1. Introduction and Problem Statement

Managing credit card benefits has become a cognitively demanding task. Modern cards offer complex, overlapping reward structures, time-sensitive offers, and enrollment requirements that consumers routinely fail to track or optimize. The result is systematic value leakage — benefits go unused, optimal cards go unchosen, and the cognitive burden of management falls entirely on the user.

This project presents an AI-assisted credit card benefits system that addresses this gap directly. The system combines LLM-powered benefit extraction, RAG-based recommendation, and human-in-the-loop validation to reduce cognitive load, surface time-sensitive opportunities, and support card selection at the point of decision.

The challenge is not only informational but also cognitive. Managing multiple credit cards requires users to remember category-specific reward rates, spending caps, enrollment requirements, offer expiration dates, and credit usage limits. This creates financial decision fatigue, where the effort required to choose the optimal card outweighs the perceived benefit. Existing personal finance tools primarily focus on budgeting and expense tracking, offering limited support for benefit awareness or reward optimization. Similarly, issuer-provided tools are typically siloed within individual card ecosystems and do not provide cross-card comparisons or integrated decision support.

Recent advances in artificial intelligence, particularly in large language models and retrieval-augmented systems, present new opportunities for building intelligent decision-support tools in personal finance. AI systems can synthesize structured rules with unstructured

information and interact with users through natural language. However, to be effective in financial contexts, such systems must balance automation with reliability, transparency, and user control.

This project develops an AI-powered personal finance assistant designed to help users understand, track, and optimize their credit card benefits. The system integrates structured benefit databases, rule-based reward calculations, and a conversational interface powered by LLMs. Users can onboard their cards, view categorized benefits, track offers, and ask natural language questions about their rewards and protections. The assistant also provides context-aware recommendations, including which card to use for a specific purchase and suggestions tailored to the user's location.

Results demonstrate that LLMs provide the greatest value as structured extraction and retrieval tools rather than autonomous decision-makers — a finding with implications for any domain where financial information is unstructured and high-stakes.

The remainder of this report is organized as follows: Section 2 reviews relevant literature; Section 3 describes the data architecture; Section 4 presents the system methods; Section 5 reports results; Sections 6 and 7 present discussion and conclusions.

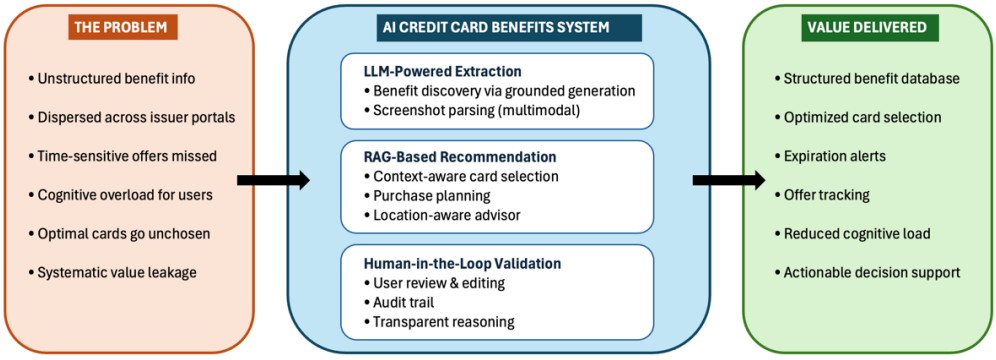

Figure 1. Conceptual Overview of the AI Credit Card Benefits System. The system bridges unstructured issuer data and actionable user decisions through three integrated AI capabilities: LLM-powered extraction, RAG-based recommendation, and human-in-the-loop validation.

2. Literature Review

The development of intelligent financial assistants lies at the intersection of personal finance technologies, recommender systems, natural language processing, and human-AI decision support. This section reviews prior work in these domains and positions of the proposed AI-powered credit card benefits assistant within the existing landscape.

2.1 Personal Finance Technologies and Decision Making

Traditional personal finance applications such as YNAB and Personal Capital have focused primarily on budgeting, expense tracking, and net worth monitoring. These tools aggregate transaction data and provide users with high-level financial visibility. However, these applications do not generally address the complex structures of modern credit cards in terms of benefits and rewards. More specialized platforms that provide rewards tracking, such as NerdWallet or Doctor Of Credit, typically offer static information about card benefits rather than personalized, context-aware guidance for individuals. As a result, a significant gap remains between available financial information and actionable decision support for credit card benefit utilization.

In 2023, the Consumer Financial Protection Bureau (CFPB) reported hundreds of consumer complaints related to credit card rewards programs. Many of these complaints involved consumers not receiving promised promotional rewards due to vague or poorly disclosed eligibility conditions. Others are concerned about the loss of benefit value as issuers modify or devalue rewards structures over time. Additionally, consumers reported difficulties

redeeming credits or benefits, often due to complex procedures and slow resolution processes when issues arose (Consumer Financial Protection Bureau 2024).

2.2 Recommender Systems and Decision Support

Recommender systems have been widely studied in domains such as e-commerce, media, and education, where they assist users in navigating large option spaces. Financial recommender systems have also emerged, particularly for portfolio allocation and product selection. However, most financial recommendation research focuses on long-term investment strategies rather than short-term, transaction-level decisions such as selecting the optimal payment method for a specific purchase.

The credit card selection problem can be viewed as a context-aware recommendation task, where the “best” option depends on variables such as purchase category, timing, reward caps, and user preferences. Traditional recommender systems often rely heavily on collaborative filtering, where recommendation is based on the preferences of similar users (Adomavicius and Tuzhilin 2005). In contrast, the credit card selection problem is largely rule-driven and therefore benefits from hybrid approaches that combine structured logic with contextual reasoning. This project builds on these ideas by integrating deterministic reward rules with AI-based natural language interaction and explanation, enabling users to receive personalized, transparent, and context-sensitive recommendations.

2.3 Natural Language Processing and Retrieval-Augmented Systems

Recent advances in large language models (LLMs) have achieved more natural and flexible interactions between users and AI systems. LLMs have been applied to question answering, document summarization, and conversational agents across various domains.

However, in knowledge-intensive tasks that require domain grounding, LLMs can produce incorrect or outdated information.

Retrieval-Augmented Generation (RAG) addresses this limitation by combining language models with external knowledge sources. Rather than relying solely on information encoded in model parameters, RAG uses a learned neural retriever to fetch relevant documents from a non-parametric memory (e.g., a dense vector index of text) and conditions the generative model on those retrieved passages to produce responses (Lewis et al. 2020). In RAG-based systems, relevant documents or structured data are retrieved in response to a user query and used to condition the model's response. This approach has shown promise in domains requiring domain-specific accuracy, including legal and medical applications. The proposed credit card benefits assistant adopts a similar paradigm, retrieving user-specific card information and benefit rules to support accurate, contextualized responses.

2.4 Human-AI Collaboration in Financial Contexts

As AI systems increasingly support financial decision-making, questions of trust, transparency, and user control become central. Research on human–AI collaboration emphasizes the importance of explainable outputs, clear communication, and mechanisms that allow users to understand and appropriately rely on system behavior (Amershi et al. 2019). Financial applications, in particular, demand higher standards of reliability and interpretability due to their direct impact on users' economic well-being.

The design of the proposed system reflects these principles by providing explanations for recommendations, distinguishing between system-derived rules and AI-generated summaries, and allowing users to explore the underlying benefits. Rather than replacing human

judgment, the assistant is intended to augment users' understanding and reduce the effort required to make informed decisions.

3. Data

The AI Credit Card Rewards Assistant integrates structured, semi-structured, and unstructured data to address a key limitation in personal finance technology: credit card benefit information is not available through standardized APIs. Instead, issuers distribute details across marketing pages, PDFs, and app interfaces, none of which provide structured, machine-readable data. To address this, the system combines user input, LLM-powered information extraction, and a relational database to build a structured model of each user's credit card portfolio.

3.1 Data Sources and Categories

The system manages five primary data categories, each originating from different sources and requiring distinct ingestion approaches.

3.1.1 Card Metadata

Basic card information (name, issuer, and annual fee) is captured via user input during card onboarding. To reduce friction and improve data consistency, the system includes a global card_templates reference table containing approximately 50 popular consumer credit cards with pre-populated metadata. When a user adds a card, they may select from this template list, which auto-fills known attributes, or manually enter details for cards not yet cataloged. Card metadata is fully structured and stored in a PostgreSQL database hosted on Neon.

3.1.2 Benefits

Benefits are the most complex category, spanning credits, lounge access, insurance, awards, and more. When a card is added, the backend uses Gemini 2.5 Flash with web grounding to retrieve benefit information and return structured JSON objects. These are validated and stored.

Benefits are classified using a project-developed taxonomy: credit, award, milestone, status, access, protection, and count_based. The milestone type was introduced to separate spend-threshold perks from immediately usable benefits, preventing misleading recommendations.

3.1.3 Offers

Issuer offers (e.g., “10% back at Whole Foods”) are time-sensitive and only visible in apps and web portals. Users upload screenshots, and Gemini’s vision-language model extracts structured fields such as merchants, discount type, value, expiration, and limits. Issuer-specific prompts improve accuracy.

3.1.4 Earning Rates

Permanent rewards multipliers (e.g., "4x on groceries") are retrieved via grounded generation and stored as canonical earning rate definitions. When a user adds a card for which earning rates already exist, the records are copied directly. Data includes category, multiplier, reward currency, estimated point value, and base vs bonus designation.

3.1.5 Usage Tracking

Users log benefit usage (e.g., Uber credit spent, lounge visits used). Data is stored in an append-only usage log, with current-period aggregations derived through queries rather than

pre-computed tables. This data powers expiration alerts and value tracking, and is the only category that is fully user-generated and requires no LLM processing.

3.1.6 Location Context

The location-aware assistant introduces a new real-time data input: the user's current geolocation. When a user requests a card suggestion based on their present location, the frontend retrieves device coordinates via the browser Geolocation API and passes them as query parameters to the backend. The backend then calls a merchant-identification API to resolve the coordinates into a merchant name and spending category. This location context is combined with the user's existing card portfolio data, including earning rates, active offers, and applicable credits, and injected into the RAG prompt. No geolocation data is persisted; it is used only at request time to support the in-the-moment recommendation.

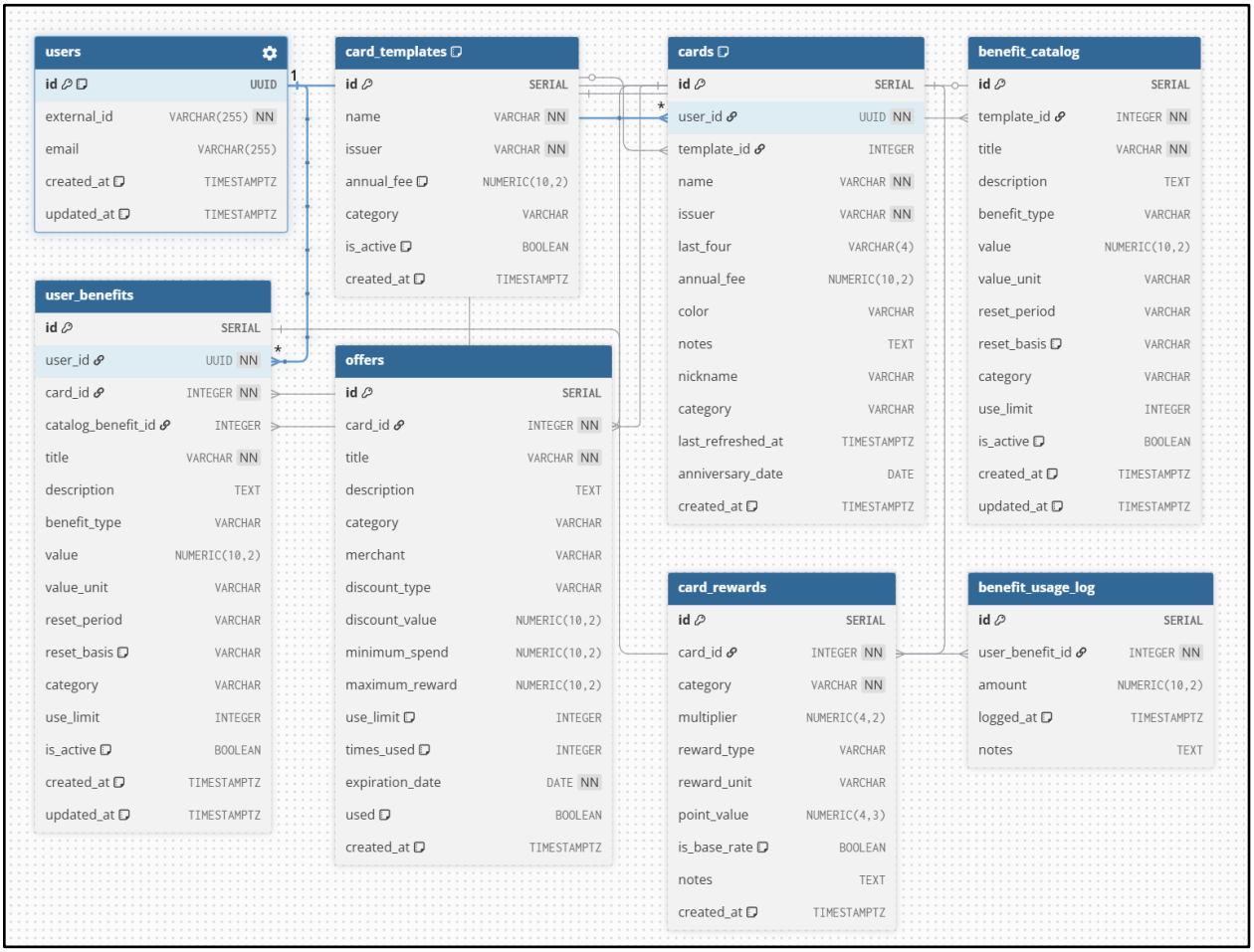

Figure 2. Entity-Relationship Diagram: The schema comprises eight tables: users, card_templates, cards, benefit_catalog, user_benefits, offers, card_rewards, and benefit_usage_log. Card templates define canonical card products, their associated benefits, and base earning rates, while user-facing tables track individual card ownership, benefit usage, and promotional offers.

3.2 Transforming Unstructured Data into Structured Data

A defining characteristic of this system is its role as a bridge between unstructured financial information and structured, queryable data. The three LLM-powered ingestion pathways, benefit discovery, offer parsing, and rewards lookup, each transform unstructured or semi-structured content into normalized relational records through structured information extraction. This transformation enables downstream features such as the dashboard, expiration

alerts, and usage tracking to operate entirely on structured data through standard database queries and aggregations.

The system's conversational features and purchase planning implement a Retrieval-Augmented Generation (RAG) architecture, where relevant records are retrieved from the application database at query time and assembled into a structured context payload that augments the LLM prompt. This grounds the model's responses in the user's actual card data rather than its parametric knowledge, reducing hallucination risk while enabling natural language understanding and semantic matching for tasks such as interpreting free-text purchase descriptions against broad spending categories.

3.3 Data Quality Challenges

Key data quality challenges include category normalization, benefit freshness, extraction errors, and subjective point valuations. Category normalization is difficult because semantically identical spending types may appear under different labels (e.g., “gas,” “gas_stations,” or “eligible_gas”), which complicates deterministic matching; a future enhancement is the introduction of a normalized category_group field. Benefit freshness is another concern, as issuer perks can change without notice and the current system performs discovery only at card creation, meaning stored data may become outdated without an automated refresh process. LLM extraction errors also occur, including occasional misclassification of benefit types, incorrect dollar values, or inconsistent date formats; while validation rules catch many common issues, edge cases remain. Finally, point valuations are inherently subjective because their real-world value depends on how rewards are redeemed in practice, so the system uses a single editable cents-per-point estimate to enable standardized comparisons while acknowledging

this simplification. To mitigate these limitations, the system incorporates validation logic and a human-in-the-loop design in which all AI-generated benefits, offers, and earning rates are editable, ensuring AI augments rather than replaces human judgment in financial decision-making.

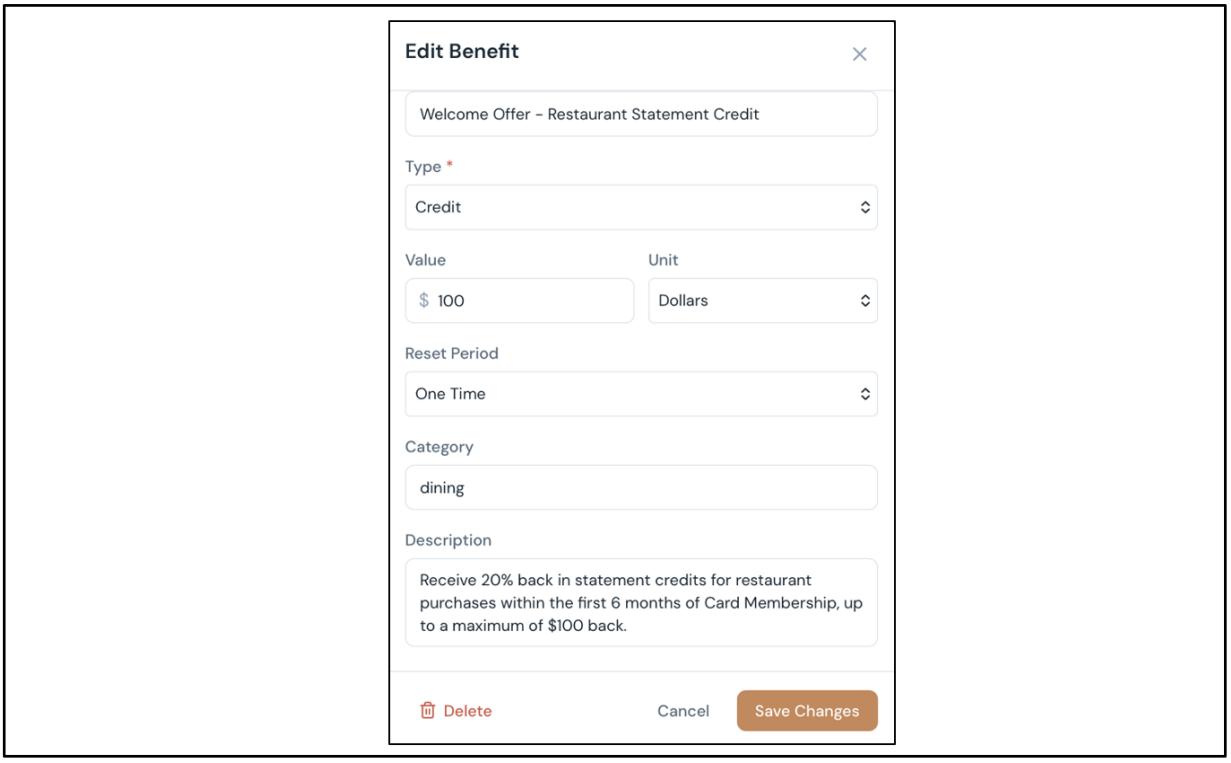

Figure 3. Benefit Editing Interface with Human-in-the-Loop Controls: The interface allows users to review and manually modify AI-extracted benefit details, including benefit type, monetary value, reset period, category, and description. This editable workflow supports data quality assurance by enabling users to correct extraction errors, adjust subjective valuations, and maintain up-to-date benefit information, reinforcing the system's human-in-the-loop design philosophy.

4. Methods

4.1 System Architecture

The system uses a client-server architecture that separates the frontend, backend, and AI services. The frontend is a Next.js single-page app, while the backend is a FastAPI REST API

that handles business logic, data access, and LLM orchestration. The frontend and backend communicate via authenticated HTTP requests. Authentication is managed by Supabase, and user isolation is enforced at the database-query level, so each request can only access the authenticated user's data.

The database is PostgreSQL hosted on Neon, chosen over the original DuckDB setup to support multi-user access. The backend uses SQLAlchemy with raw SQL for flexible query control. LLM functionality is powered by Gemini 2.5 Flash, selected for cost efficiency, vision capabilities, and web-grounded retrieval. An abstraction layer allows the system to switch LLM providers without changing application logic.

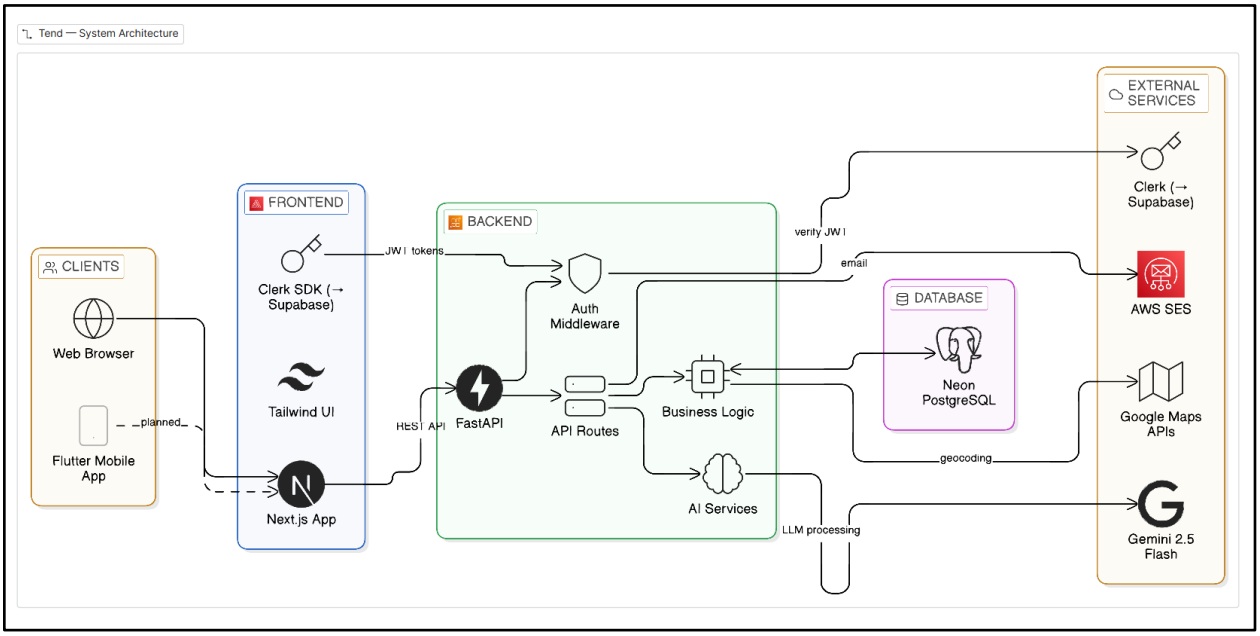

Figure 4. System Architecture: The application uses a Next.js frontend, FastAPI backend, and Neon PostgreSQL database, with Google Gemini 2.5 Flash providing AI capabilities. Business logic and AI services are separated to distinguish deterministic operations from LLM-dependent tasks.

4.2 Frontend Architecture

The frontend uses Next.js page routing with domain-based components (cards, dashboard, modals). Data fetching uses SWR with custom hooks that inject Supabase authentication tokens automatically.

Components follow a controlled architecture pattern, where parent components own state and modals act as presentation-only components. This avoids double-render and synchronization bugs. The dashboard aggregates benefits, offers, and usage data into a unified “Needs Attention” view.

4.3 API Design

The backend exposes a resource-based REST API for cards, benefits, offers, rewards, usage, dashboard, and chat. Pydantic models enforce strict validation, reject unexpected fields, and serialize responses with computed fields to reduce frontend round trips.

Error handling follows HTTP conventions: 404 for missing resources, 400 for validation failures, and 500 for unexpected server errors. FastAPI's automatic OpenAPI documentation provides an interactive API reference that was used throughout development for manual endpoint testing alongside the automated test suite.

Domain	Endpoints	Description
Cards	GET, POST `/cards` GET, PUT, DELETE `/cards/{id}` POST `/cards/{id}/reload`	CRUD operations and LLM-powered benefit/reward reload
Benefits	GET, POST `/cards/{id}/benefits` PUT, DELETE `/benefits/{id}` POST `/benefits/{id}/log-credit` POST `/benefits/{id}/log-visit`	CRUD and usage logging (credits, visits)
Rewards	GET `/cards/{id}/rewards`	Earning rates per card
Offers	GET, POST `/cards/{id}/offers` POST `/cards/{id}/offers/upload` DELETE `/offers/{id}`	CRUD and screenshot-based offer parsing
Dashboard	GET `/dashboard/summary`	Aggregated metrics and attention items
Chat	POST `/chat`	RAG-based conversational advisor
Purchase Planning	POST `/purchases/plan`	Card recommendation for planned transactions
Recommendations	GET `/recommendations/nearby`	Location-based card suggestions

Figure 5. REST API Endpoints by Resource Domain. The backend exposes 21 endpoints across eight domains, supporting card management, benefit and offer tracking, usage logging, and AI-powered features including conversational advising, purchase planning, and screenshot-based offer extraction.

4.4 Development Methodology

The project used a lightweight Agile approach with one-week sprints. Responsibilities were divided by strengths: product management and research on one side, full-stack engineering on the other. Features were developed iteratively from specification → backend → frontend → integration.

Code quality is maintained through Ruff (Python) and ESLint (TypeScript). Dependency management uses uv for Python and npm for frontend.

4.5 Testing Strategy

The system is supported by a comprehensive test suite of over 130 tests spanning both unit and integration levels. Unit tests isolate components by mocking external dependencies, allowing verification of CRUD logic and schema validation in a controlled environment. Integration tests use an in-memory SQLite database to simulate full database lifecycles, ensuring that data operations behave correctly across realistic workflows. LLM-powered features are tested using mocked API calls, which makes it possible to validate prompt construction, response parsing, and error handling without consuming API quota.

4.6 Card Onboarding and Benefit Extraction

When a user adds a credit card, the system launches an automated onboarding pipeline. The user either selects a card from a reference template or enters details manually, after which the backend checks whether canonical benefit and reward data already exists for that card product. If so, the records are copied directly to the user's account. If not, the backend triggers two LLM-powered extraction steps.

First, a benefit lookup service uses grounded generation to retrieve the card’s current benefits from live web sources and returns structured entries aligned with the system’s benefit taxonomy. These results are validated, normalized, and stored in the database.

Second, a rewards lookup service applies the same approach to identify category-based earning rates, including multipliers and estimated point valuations. This process reduces the need for users to manually research and enter card details, while still allowing them to review and edit AI-generated data through the frontend (see Appendix Figure A1).

4.7 Offer Detection via Screenshot Parsing

Promotional offers appear only as visual elements in issuer apps and portals, so the system uses a multimodal pipeline to extract structured data from screenshots. Users upload an offers-page screenshot, which the backend sends to Gemini 2.5 Flash with a prompt that asks the model to return each offer in JSON format, including merchant, discount type and value, minimum spend, expiration date, and usage limits. To improve accuracy, the system applies issuer-specific prompt templates (e.g., for American Express and Chase) that account for different layout and wording styles. Extracted offers are validated, converted into structured records, and saved to the database, where users can review, edit, or delete them through the frontend (see Appendix Figure A2).

4.8 Conversational Advisor

The system includes a chatbot that supports three main tasks: recommending the best card for a merchant, browsing active offers and benefits, and identifying upcoming expirations. It uses a Retrieval-Augmented Generation (RAG) approach, where the backend classifies user intent and retrieves relevant data such as earning rates, active offers, and expiring benefits from the user's database. This structured context is added to the prompt before sending it to Gemini, ensuring responses are grounded in the user's actual card information and reducing hallucination risk. To maintain reliability, the initial version is limited to single-turn interactions and does not yet support complex comparisons or advanced financial analysis.

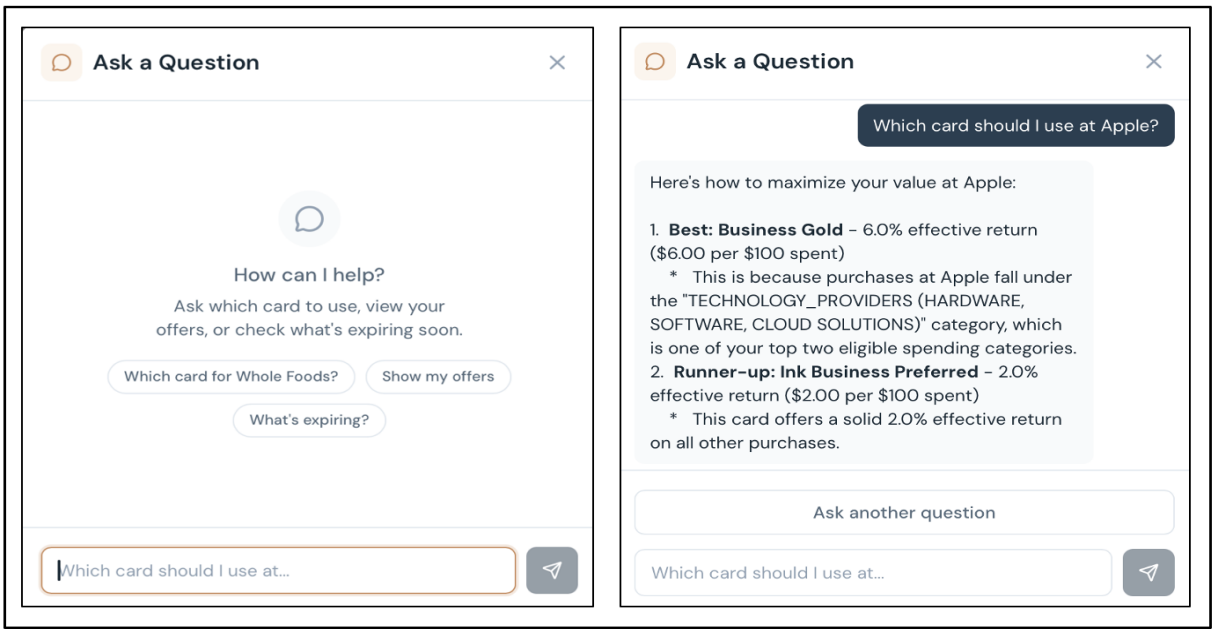

Figure 6: RAG-Based Conversational Advisor for Merchant Optimization: The chatbot generates a recommendation by retrieving user-specific card data, including earning rates and reward structures, and injecting this context into the LLM prompt. The resulting response provides explainable, grounded guidance tailored to the user's portfolio.

4.9 Purchase Planning

The purchase planning feature lets users enter a planned transaction, including amount, category, date, and an optional description, to receive a recommendation for the most valuable card to use. It follows a RAG approach: the backend retrieves relevant card data and provides it as context to the LLM. The model semantically matches the purchase description to earning categories and reasons across offers, credits, and rewards to rank cards by value. An LLM is used because free-text descriptions and inconsistent category labels are difficult to handle with rigid rules, and because combining multiple benefit types would require complex logic. The system returns a structured response with the best card, a value breakdown, and ranked alternatives.

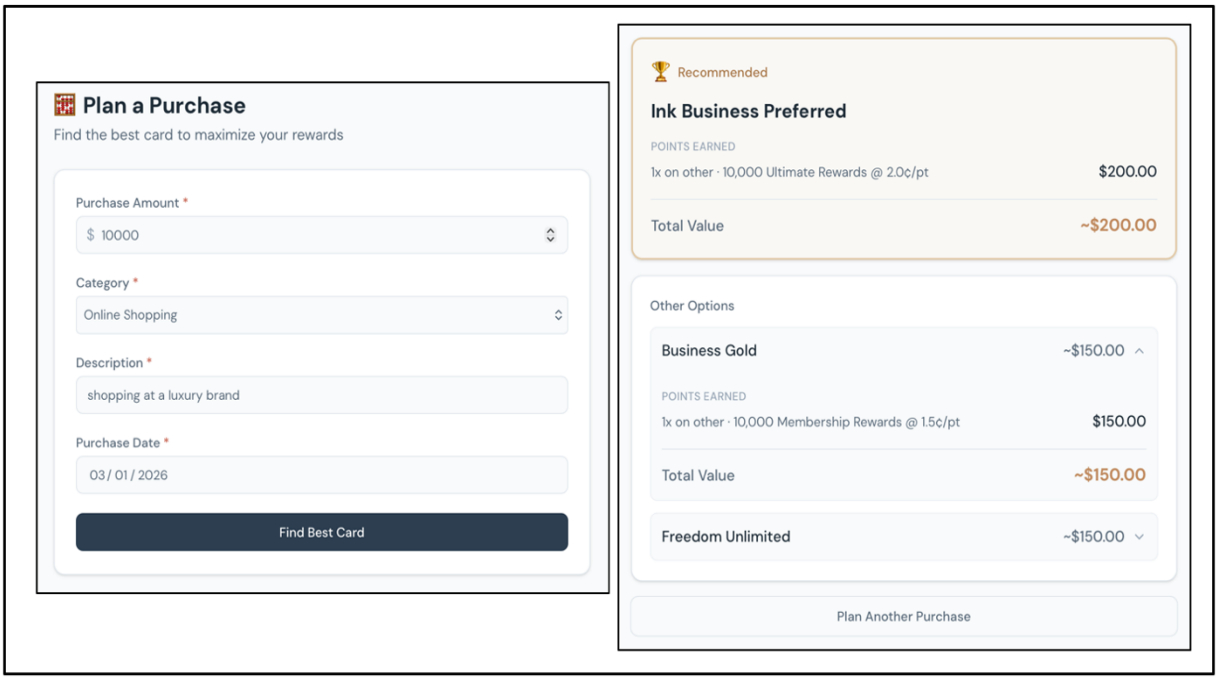

Figure 7. RAG-Based Purchase Planning and Card Ranking: The system analyzes a planned transaction using retrieved card metadata and LLM-based semantic matching to estimate total reward value. The output presents the recommended card, a value breakdown, and ordered alternatives to support informed payment decisions.

4.10 Location-Aware Assistant

The location-aware assistant extends the RAG-based recommendation pipeline with real-time geospatial context. When a user opens the feature on mobile or in a browser that supports the Geolocation API, the frontend requests device coordinates and passes them to a dedicated backend endpoint. The backend resolves the coordinates to a merchant name and spending category using a merchant-identification API and constructs a context payload containing the inferred merchant, category, and the user's relevant card data such as earning rates, active offers, and applicable credits. This payload is injected into the LLM prompt alongside an instruction to return the card that maximizes reward value for the detected

merchant. The response follows the same structured format used by the purchase planner, including a recommended card, value breakdown, and ranked alternatives.

To preserve privacy, geolocation data is used only at request time and is not stored in the database. Merchant resolution is performed server-side so that raw coordinates are not exposed to the LLM. Fallback behavior is provided when location permission is denied, or a merchant cannot be resolved: the assistant surfaces a manual input form that mirrors the standard purchase planner. This design ensures the feature degrades gracefully across devices and permission contexts.

4.11 Prompt Engineering

All LLM prompts are stored as external files in a dedicated prompts/ directory, separate from application logic. This design allows prompts to be refined, versioned, and audited without requiring code changes. Prompt content evolved iteratively as new edge cases emerged during development and testing.

The coding process for this project followed an AI-assisted, human-in-the-loop workflow. Rather than writing all components from scratch, generative AI tools were used to accelerate prototyping, debugging, and iterative refinement. Initial prompts were used to generate baseline code structures for key modules (e.g., data preprocessing, retrieval logic, and recommendation functions). These outputs were then manually reviewed, tested, and modified to align with project requirements and domain constraints.

The development process was inherently iterative. Prompt instructions were progressively refined based on observed model outputs, runtime behavior, and performance evaluation. In many cases, AI-generated code served as a starting scaffold that was

subsequently optimized for efficiency, correctness, and integration with the broader system architecture. Human oversight remained central throughout the process, including validation of logic, handling of edge cases, and alignment with financial domain rules.

Several targeted prompt refinements were introduced over time. For example, explicit instructions were added to distinguish immediate benefits from spending-threshold milestones, and issuer-specific templates were developed to improve screenshot parsing accuracy. Observed model behavior and data quality checks informed successive prompt updates.

Each prompt requests structured JSON output that conforms to predefined schemas. A deterministic parsing layer then cleans, validates, and normalizes responses before persisting them to the database. Schema validation rules and automated quality checks are applied to detect malformed or incomplete responses prior to acceptance. This combination of structured prompt design, AI-assisted iterative development, and deterministic post-processing ensures that LLM outputs are transformed into reliable, typed records suitable for downstream recommendation and analytics workflows.

Importantly, all final code used in the system reflects human validation and customization beyond the initial AI-generated suggestions.

5. Results

5.1 Benefit Discovery

The benefit discovery pipeline uses grounded generation to retrieve and structure card benefits from live web sources. In testing, a single LLM call typically identified most benefits for premium cards (approximately 12–18 items). The primary error observed was type conflation,

where spending-threshold rewards were misclassified as immediately available benefits. Introducing a distinct milestone benefit type and refining prompt instructions significantly reduced this issue. Other errors included value misattribution, fabricated benefits, and omissions. While validation rules filter malformed entries, factually incorrect but well-formatted data may still pass, reinforcing the system's human-in-the-loop design where users review and edit generated records (see Appendix Figure A3).

5.2 Offer Extraction from Screenshots

The multimodal screenshot pipeline reliably extracted core offer attributes (merchant, discount type, value) from issuer portals. Accuracy varied by issuer, with American Express layouts performing best and Chase requiring a specialized prompt template. Common errors involved expiration dates, minimum spend conditions, and usage limits embedded in fine print. A multi-format date parser and validation rules reduced ingestion errors, and rejected offers are surfaced to users for manual correction.

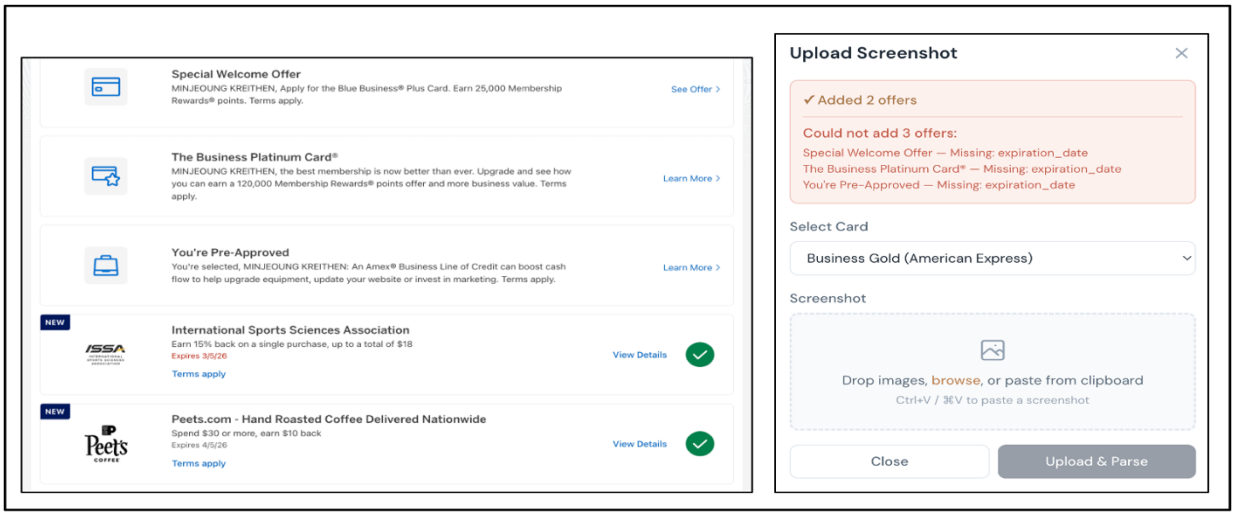

Figure 8. Before-and-After Offer Extraction with Error Handling: The system processes an issuer screenshot to produce structured offer records while surfacing validation failures for incomplete fields. This workflow demonstrates both extraction capability and built-in safeguards for data quality.

5.3 Earning Rate Discovery

Grounded generation successfully identified major bonus earning categories for tested cards. However, inconsistent category labels across cards created an entity resolution challenge (e.g., multiple surface forms for fuel purchases). Rather than enforcing static normalization, the system currently includes all earning rates in the RAG context and relies on semantic matching, with deterministic normalization planned for future work. Additionally, raw multipliers proved insufficient for cross-program comparison, leading to the introduction of effective return calculations based on stored point valuations.

5.4 Card Recommendations

RAG-based recommendation features produced accurate results when underlying earning rates and valuations were complete. The system correctly selected cards with the highest effective return while incorporating relevant offers and credits. Structured value breakdowns improved transparency and user trust, allowing verification and informed trade-offs. Recommendation quality decreased when card data was incomplete, highlighting the importance of accurate upstream extraction (see Appendix Figure A4).

5.5 Dashboard and Expiration Tracking

The dashboard effectively surfaced benefits and offers expiring within 30 days and tracked benefit usage over time. A backend issue initially prevented offers from appearing in expiration alerts, but it was resolved, restoring correct grouping and sorting. Usage logging created a feedback loop by encouraging benefit utilization and recording realized value.

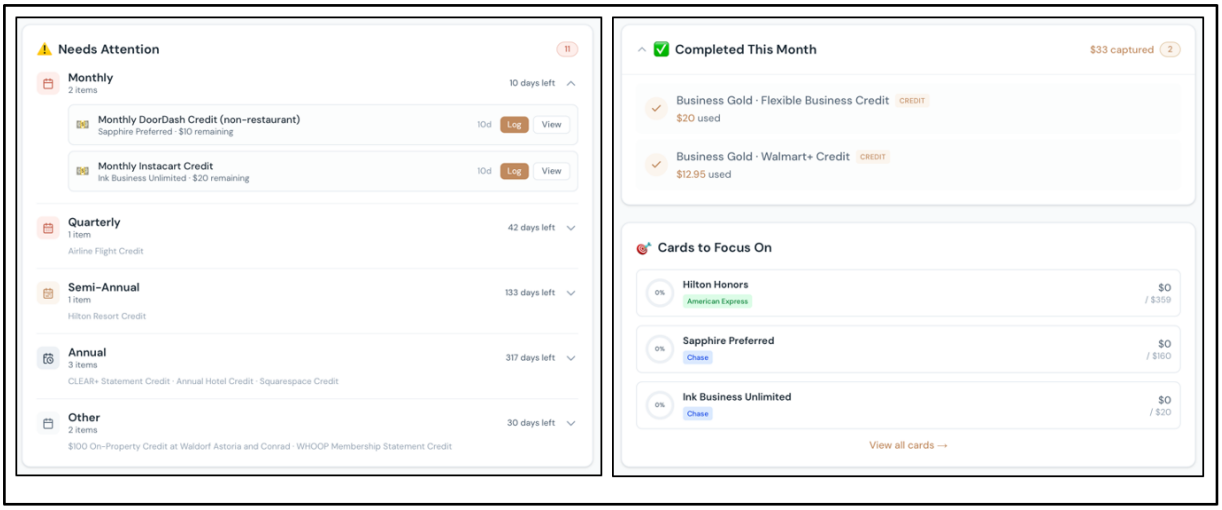

Figure 9. Expiration Tracking and Benefit Utilization Dashboard: The interface surfaces time-sensitive benefits and summarizes recent usage activity, demonstrating the system's ability to support proactive benefit management and value tracking.

5.6 Location-Aware Assistant

The location-aware assistant performed reliably across the tested scenarios. Merchant resolution succeeded for well-known chain retailers and restaurants, correctly mapping coordinates to recognizable merchant names and standard spending categories such as dining, groceries, and travel. In these cases, the downstream RAG recommendation matched or outperformed the manual purchase planner in response time, since no user input beyond location consent was required. Card recommendations were consistent with those produced by the standard purchase planner when the same merchant and category were supplied manually, confirming that the pipeline correctly integrates geolocation context without altering recommendation logic.

Merchant resolution was less reliable for independent or unfamiliar businesses, where the API returned generic categories (e.g., "retail store") rather than a specific merchant name. In these cases, the LLM still returned a recommendation, but the value breakdown was based

on the broad category rather than any merchant-specific offers, reducing precision compared to manually specified inputs. The graceful fallback to manual entry was triggered correctly when location permission was denied or when coordinates could not be resolved to a merchant, ensuring the feature did not block the user's workflow. Overall, the location-aware assistant successfully transitioned the tool from a planning aid into an in-the-moment decision companion for common retail and dining contexts.

6. Discussion

6.1 AI as Augmentation, Not Replacement

A key finding is that AI provides the most value as an information extraction and retrieval layer rather than as a decision-making engine. LLMs convert unstructured sources into structured, user-verifiable data, while core system features such as dashboards, alerts, usage tracking, and value calculations operate through deterministic database logic. The purchase planner illustrates this balance: mathematical optimization is rule-based, but the LLM is used for semantic interpretation of free-text purchase descriptions and inconsistent category labels.

6.2 Interpreting Accuracy in a Human-in-the-Loop System

Since all AI-generated records are editable, system effectiveness depends less on perfect accuracy and more on net time saved relative to manual entry. Most improvements during development came from prompt design, clearer taxonomies, and validation rules rather than model changes, indicating that schema and prompt engineering are high-leverage interventions for domain-specific extraction tasks.

6.3 Entity Resolution and the Limits of Semantic Matching

Inconsistent category labels across cards present an entity resolution challenge. The system currently relies on LLM semantic matching within RAG contexts, while a future normalized category layer would enable deterministic comparisons without losing descriptive richness.

6.4 Point Valuation and its Subjectivity

Effective return calculations require estimated point values, which are inherently subjective. Allowing users to edit these values balances usability with personalization while avoiding the complexity of modeling multiple redemption scenarios.

6.5 Risks and Limitations

6.5.1 Data Staleness

Benefit and earning rate information is populated through grounded generation at card creation time and does not automatically refresh. Issuers frequently modify benefit packages by adding, removing, or restructuring perks, sometimes without prominent notification. A user who added their card months ago may have stale benefit data without realizing it. The reload endpoint allows manual refresh, but automated change detection remains outside the current scope.

6.5.2 Hallucination in Financial Contexts

While the RAG-based chatbot and purchase planner are grounded in the user's actual card data, which is retrieved from the application database and injected into the LLM context at query time, the benefit and rewards lookup features rely on the model's web search capabilities during grounded generation, which can produce plausible but incorrect

information. In a financial tool, even small inaccuracies can erode user trust. The system mitigates this through transparency by ensuring benefits and earning rates are displayed in full detail, allowing users to verify entries, but cannot fully eliminate the risk. This is a well-documented limitation of generative AI systems, and the human-in-the-loop design described in the Data section represents the project's primary defense.

6.5.3 Scope Constraints

The chatbot is deliberately limited to three use cases with single-turn interactions, and the system does not support multi-turn conversational state, complex benefit term interpretation, or insurance claims assistance. These constraints were imposed to deliver a reliable initial RAG implementation, but they mean the system cannot yet address some of the highest-value use cases in the credit card optimization space, particularly claims filing, which emerged during product brainstorming as a significant unmet need.

7. Conclusions

This project addresses a gap in personal finance technology by building an AI-assisted system for understanding, tracking, and optimizing credit card benefits. The system combines LLM-powered extraction, a structured database, and RAG-based recommendations to reduce cognitive load in everyday financial decisions.

A central conclusion is that AI is most effective in this domain when used to structure information rather than replace deterministic logic. Many performance gains arose from improved prompt structure and schema design rather than model changes, underscoring the

importance of domain modeling. The human-in-the-loop approach further reframes success as time saved with user oversight rather than full automation.

Overall, the work demonstrates that grounded, retrieval-based AI can provide transparent and practical financial decision support, while highlighting future opportunities such as automated benefit refresh and more advanced advisory capabilities.

8. Directions for Future Work

Future work focuses on expanding data automation, recommendation sophistication, contextual awareness, and personalization.

A key priority is automating offer ingestion. Replacing manual screenshot uploads with direct retrieval from issuer portals would reduce user friction, though technical and legal barriers such as bot detection and anti-scraping protections pose challenges. Interim solutions like browser extensions or notification-based triggers may offer practical compromises.

Another major extension is multi-card optimization for large purchases. Instead of recommending a single card, the system could allocate spending across multiple cards to maximize combined rewards, credits, and offers. This would require solving a constrained optimization problem and extending LLM reasoning to produce sequenced, multi-card recommendations.

Building on the location-aware assistant now in production, further geospatial enhancements could improve merchant resolution accuracy for independent and small businesses, where generic category fallbacks currently reduce recommendation precision.

Richer merchant databases, user-contributed corrections, and category disambiguation layers would strengthen the feature's reliability across a broader range of retail contexts.

Finally, the system's usage data enables predictive, personalized advising. Longitudinal analysis of benefit usage could identify underutilized perks, detect changes in spending behavior, and estimate whether a card's annual fee remains justified. This would shift the system from reactive reminders to proactive portfolio optimization. To support these capabilities in real-world settings, future development will also deliver the platform as native mobile applications for iOS and Android, enabling real-time, on-the-go decision support and deeper integration with location and notification services.

Data Availability

The dataset used in this project consists of user-entered card metadata, AI-extracted benefit and offer records, and usage logs generated during system testing. No proprietary or third-party datasets were used. The schema and sample data are available from the corresponding author upon reasonable request.

Code Availability

The full source code for the AI Credit Card Rewards Assistant, including the FastAPI backend, Next.js frontend, prompt files, and test suite, is available from the corresponding author upon reasonable request.

Use of AI in Developing This Work

Generative AI tools were used throughout the development of this application in an AI-assisted, human-in-the-loop workflow. Claude was used as the primary coding assistant for

prototyping, debugging, and iterative refinement across the full stack, including FastAPI backend routes, SQLAlchemy database operations, Next.js frontend components, and test suite development. Initial prompts generated baseline code structures that were then reviewed, tested, and customized to meet project requirements. Prompt instructions were progressively refined based on observed behavior, runtime errors, and performance evaluation. In many cases, AI-generated code served as a starting scaffold that was subsequently optimized for correctness, maintainability, and alignment with the project's architectural patterns. All final code reflects human validation, and no AI-generated output was deployed without review and testing.

References

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Appendix A

Figure A1. Card onboarding flow.
Figure A1 shows the card onboarding process, including template selection, automated benefit population, reward retrieval, and user review. The workflow combines deterministic template reuse with LLM-based discovery and concludes with user verification to ensure data quality.

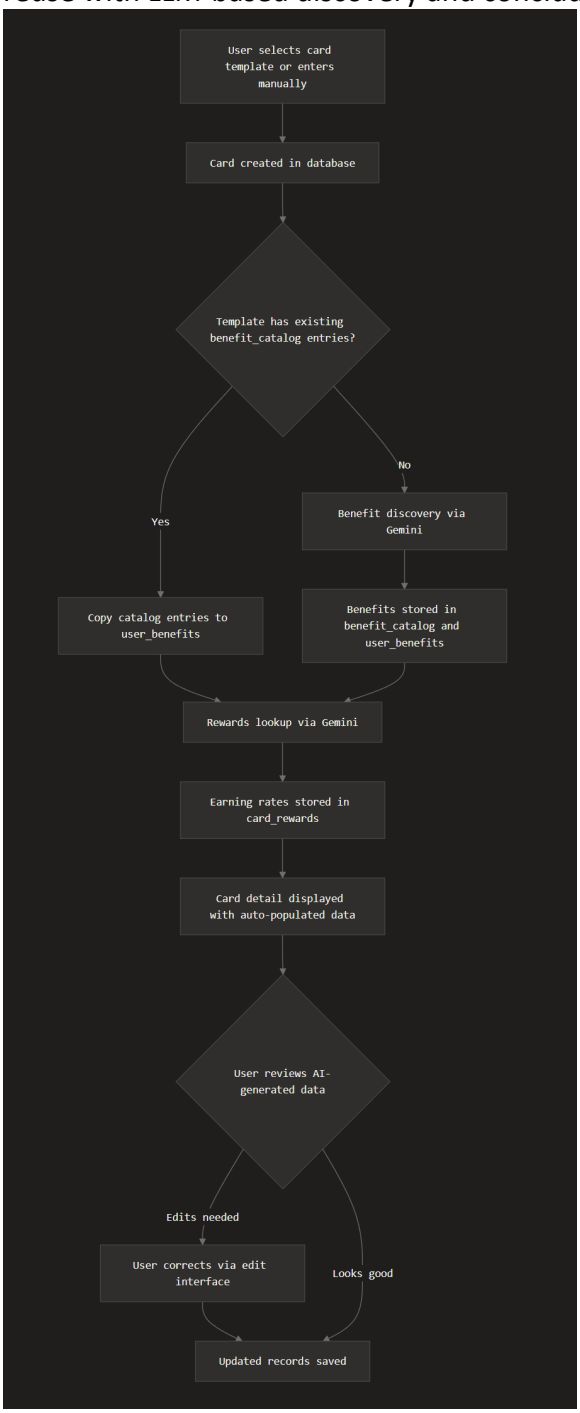


Figure A2. Structured Offer Extraction from Issuer Screenshot.
The interface displays promotional offers after multimodal parsing of an issuer offers-page screenshot. Each offer is converted into a structured record containing merchant, reward value, spending requirement, and expiration metadata. Users can review and edit extracted offers in the frontend, supporting validation and correction of AI-generated information.

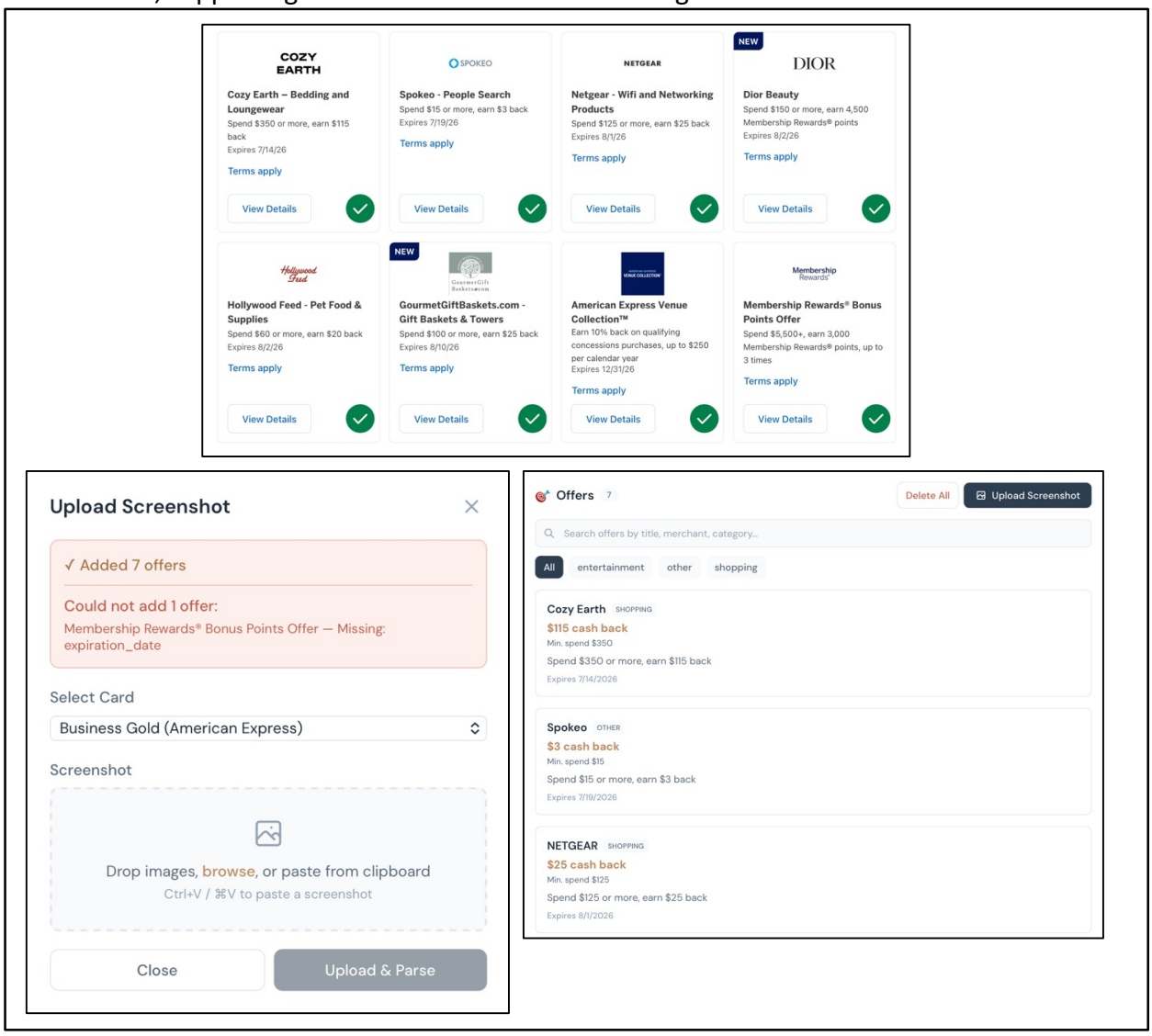
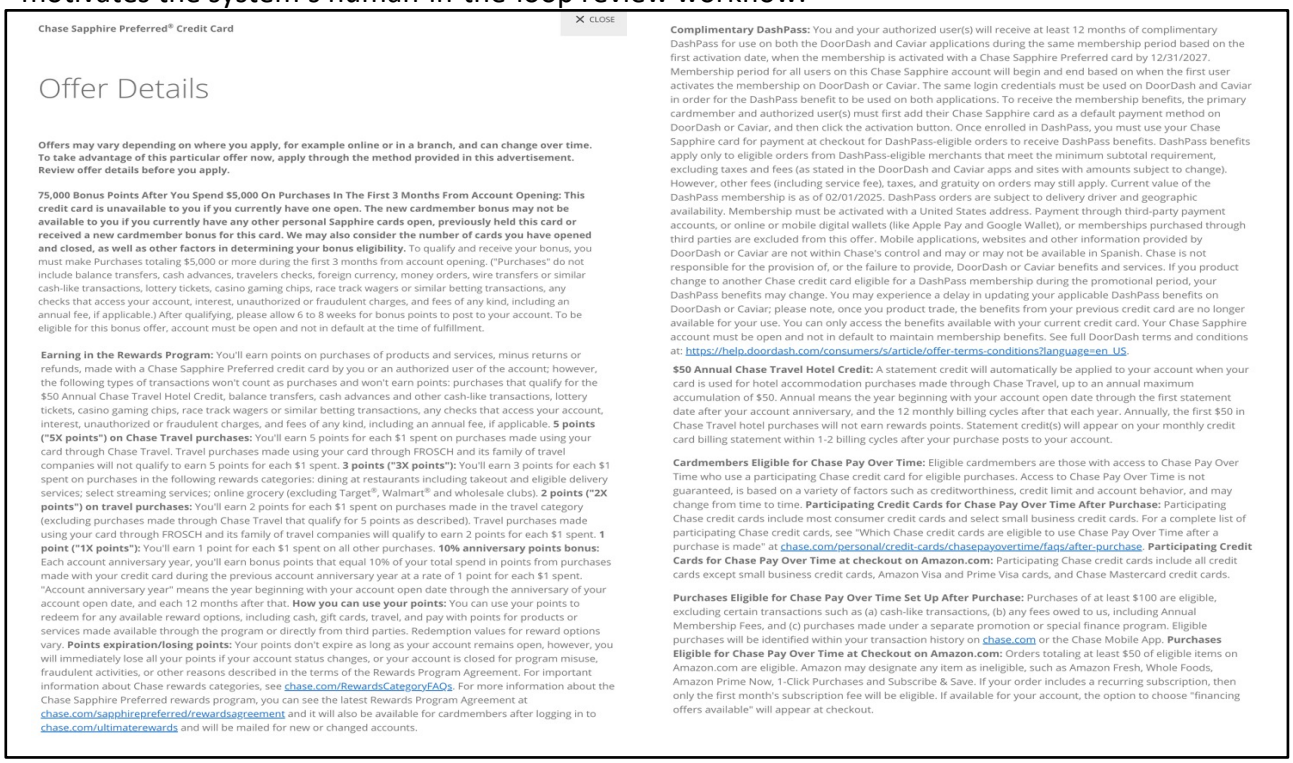


Figure A3. Benefit Discovery Output Compared with Issuer-Published Benefits.
The figure illustrates LLM-generated benefit extraction for a premium card alongside the issuer's published benefit list. Most core benefits are correctly identified, while remaining discrepancies include occasional omissions and type misclassifications. This comparison highlights both the effectiveness and the limitations of the grounded discovery pipeline and motivates the system's human-in-the-loop review workflow.



Chase Sapphire Preferred® Credit Card

✕ CLOSE

Offer Details

Offers may vary depending on where you apply, for example online or in a branch, and can change over time. To take advantage of this particular offer now, apply through the method provided in this advertisement. Review offer details before you apply.

75,000 Bonus Points After You Spend $5,000 On Purchases In The First 3 Months From Account Opening: This credit card is unavailable to you if you currently have one open. The new cardmember bonus may not be available to you if you currently have any other personal Sapphire cards open, previously held this card or received a new cardmember bonus for this card. We may also consider the number of cards you have opened and closed, as well as other factors in determining your bonus eligibility. To qualify and receive your bonus, you must make Purchases totaling $5,000 or more during the first 3 months from account opening. ("Purchases" do not include balance transfers, cash advances, travelers checks, foreign currency, money orders, wire transfers or similar cash-like transactions, lottery tickets, casino gaming chips, race track wagers or similar betting transactions, any checks that access your account, interest, unauthorized or fraudulent charges, and fees of any kind, including an annual fee, if applicable.) After qualifying, please allow 6 to 8 weeks for bonus points to post to your account. To be eligible for this bonus offer, account must be open and not in default at the time of fulfillment.

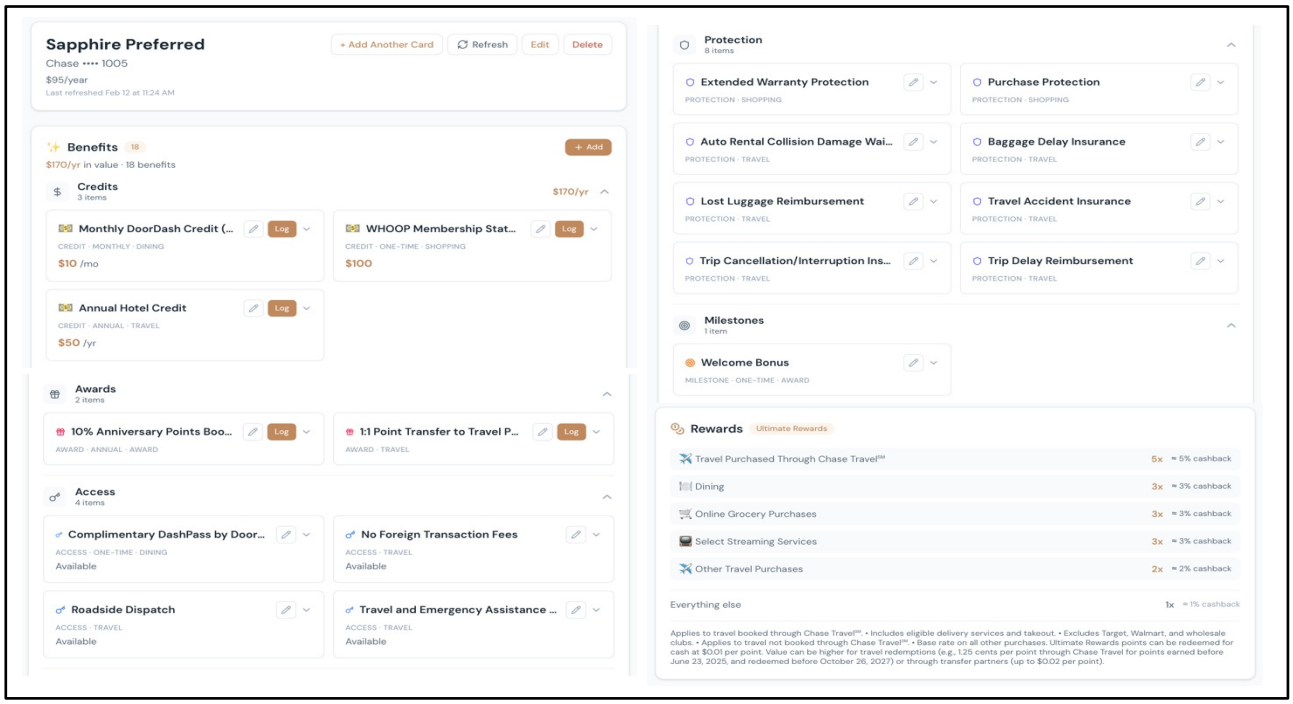
Earning in the Rewards Program: You'll earn points on purchases of products and services, minus returns or refunds, made with a Chase Sapphire Preferred credit card by you or an authorized user of the account; however, the following types of transactions won't count as purchases and won't earn points: purchases that qualify for the $50 Annual Chase Travel Hotel Credit, balance transfers, cash advances and other cash-like transactions, lottery tickets, casino gaming chips, race track wagers or similar betting transactions, any checks that access your account, interest, unauthorized or fraudulent charges, and fees of any kind, including an annual fee, if applicable. **5 points ("5X points") on Chase Travel purchases:** You'll earn 5 points for each $1 spent on purchases made using your card through Chase Travel. Travel purchases made using your card through FROSCH and its family of travel companies will not qualify to earn 5 points for each $1 spent. **3 points ("3X points"):** You'll earn 3 points for each $1 spent on purchases in the following rewards categories: dining at restaurants including takeout and eligible delivery services; select streaming services; online grocery (excluding Target®, Walmart® and wholesale clubs). **2 points ("2X points") on travel purchases:** You'll earn 2 points for each $1 spent on purchases made in the travel category (excluding purchases made through Chase Travel that qualify for 5 points as described). Travel purchases made using your card through FROSCH and its family of travel companies will qualify to earn 2 points for each $1 spent. **1 point ("1X points"):** You'll earn 1 point for each $1 spent on all other purchases. **10% anniversary points bonus:** Each account anniversary year, you'll earn bonus points that equal 10% of your total spend in points from purchases made with your credit card during the previous account anniversary year at a rate of 1 point for each $1 spent. "Account anniversary year" means the year beginning with your account open date through the anniversary of your account open date, and each 12 months after that. **How you can use your points:** You can use your points to redeem for any available reward options, including cash, gift cards, travel, and pay with points for products or services made available through the program or directly from third parties. Redemption values for reward options vary. **Points expiration/losing points:** Your points don't expire as long as your account remains open, however, you will immediately lose all your points if your account status changes, or your account is closed for program misuse, fraudulent activities, or other reasons described in the terms of the Rewards Program Agreement. For important information about Chase rewards categories, see chase.com/RewardsCategoryFAQs. For more information about the Chase Sapphire Preferred rewards program, you can see the latest Rewards Program Agreement at chase.com/sapphirepreferred/rewardsagreement and it will also be available for cardmembers after logging in to chase.com/ultimaterewards and will be mailed for new or changed accounts.

Complimentary DashPass: You and your authorized user(s) will receive at least 12 months of complimentary DashPass for use on both the DoorDash and Caviar applications during the same membership period based on the first activation date, when the membership is activated with a Chase Sapphire Preferred card by 12/31/2027. Membership period for all users on this Chase Sapphire account will begin and end based on when the first user activates the membership on DoorDash or Caviar. The same login credentials must be used on DoorDash and Caviar in order for the DashPass benefit to be used on both applications. To receive the membership benefits, the primary cardmember and authorized user(s) must first add their Chase Sapphire card as a default payment method on DoorDash or Caviar, and then click the activation button. Once enrolled in DashPass, you must use your Chase Sapphire card for payment at checkout for DashPass-eligible orders to receive DashPass benefits. DashPass benefits apply only to eligible orders from DashPass-eligible merchants that meet the minimum subtotal requirement, excluding taxes and fees (as stated in the DoorDash and Caviar apps and sites with amounts subject to change). However, other fees (including service fee), taxes, and gratuity on orders may still apply. Current value of the DashPass membership is as of 02/01/2025. DashPass orders are subject to delivery driver and geographic availability. Membership must be activated with a United States address. Payment through third-party payment accounts, or online or mobile digital wallets (like Apple Pay and Google Wallet), or memberships purchased through third parties are excluded from this offer. Mobile applications, websites and other information provided by DoorDash or Caviar are not within Chase's control and may or may not be available in Spanish. Chase is not responsible for the provision of, or the failure to provide, DoorDash or Caviar benefits and services. If you product change to another Chase credit card eligible for a DashPass membership during the promotional period, your DashPass benefits may change. You may experience a delay in updating your applicable DashPass benefits on DoorDash or Caviar; please note, once you product trade, the benefits from your previous credit card are no longer available for your use. You can only access the benefits available with your current credit card. Your Chase Sapphire account must be open and not in default to maintain membership benefits. See full DoorDash terms and conditions at: https://help.doordash.com/consumers/s/article/offer-terms-conditions?language=en_US.

$50 Annual Chase Travel Hotel Credit: A statement credit will automatically be applied to your account when your card is used for hotel accommodation purchases made through Chase Travel, up to an annual maximum accumulation of $50. Annual means the year beginning with your account open date through the first statement date after your account anniversary, and the 12 monthly billing cycles after that each year. Annually, the first $50 in Chase Travel hotel purchases will not earn rewards points. Statement credit(s) will appear on your monthly credit card billing statement within 1-2 billing cycles after your purchase posts to your account.

Cardmembers Eligible for Chase Pay Over Time: Eligible cardmembers are those with access to Chase Pay Over Time who use a participating Chase credit card for eligible purchases. Access to Chase Pay Over Time is not guaranteed, is based on a variety of factors such as creditworthiness, credit limit and account behavior, and may change from time to time. **Participating Credit Cards for Chase Pay Over Time After Purchase:** Participating Chase credit cards include most consumer credit cards and select small business credit cards. For a complete list of participating Chase credit cards, see "Which Chase credit cards are eligible to use Chase Pay Over Time after a purchase is made" at chase.com/personal/credit-cards/chasepayovertime/faqs/after-purchase. **Participating Credit Cards for Chase Pay Over Time at checkout on Amazon.com:** Participating Chase credit cards include all credit cards except small business credit cards, Amazon Visa and Prime Visa cards, and Chase Mastercard credit cards.

Purchases Eligible for Chase Pay Over Time Set Up After Purchase: Purchases of at least $100 are eligible, excluding certain transactions such as (a) cash-like transactions, (b) any fees owed to us, including Annual Membership Fees, and (c) purchases made under a separate promotion or special finance program. Eligible purchases will be identified within your transaction history on chase.com or the Chase Mobile App. **Purchases Eligible for Chase Pay Over Time at Checkout on Amazon.com:** Orders totaling at least $50 of eligible items on Amazon.com are eligible. Amazon may designate any item as ineligible, such as Amazon Fresh, Whole Foods, Amazon Prime Now, 1-Click Purchases and Subscribe & Save. If your order includes a recurring subscription, then only the first month's subscription fee will be eligible. If available for your account, the option to choose "financing offers available" will appear at checkout.



Sapphire Preferred
Chase •••• 1005
$95/year
Last refreshed Feb 12 at 11:24 AM

+ Add Another Card | Refresh | Edit | Delete

Benefits 18 — + Add
$170/yr in value · 18 benefits

Credits 3 items — $170/yr
- Monthly DoorDash Credit (... — CREDIT · MONTHLY · DINING — $10 /mo — Log
- WHOOP Membership Stat... — CREDIT · ONE-TIME · SHOPPING — $100 — Log
- Annual Hotel Credit — CREDIT · ANNUAL · TRAVEL — $50 /yr — Log

Awards 2 items
- 10% Anniversary Points Boo... — AWARD · ANNUAL · AWARD — Log
- 1:1 Point Transfer to Travel P... — AWARD · TRAVEL — Log

Access 4 items
- Complimentary DashPass by Door... — ACCESS · ONE-TIME · DINING — Available
- No Foreign Transaction Fees — ACCESS · TRAVEL — Available
- Roadside Dispatch — ACCESS · TRAVEL — Available
- Travel and Emergency Assistance ... — ACCESS · TRAVEL — Available

Protection 8 items
- Extended Warranty Protection — PROTECTION · SHOPPING
- Purchase Protection — PROTECTION · SHOPPING
- Auto Rental Collision Damage Wai... — PROTECTION · TRAVEL
- Baggage Delay Insurance — PROTECTION · TRAVEL
- Lost Luggage Reimbursement — PROTECTION · TRAVEL
- Travel Accident Insurance — PROTECTION · TRAVEL
- Trip Cancellation/Interruption Ins... — PROTECTION · TRAVEL
- Trip Delay Reimbursement — PROTECTION · TRAVEL

Milestones 1 item
- Welcome Bonus — MILESTONE · ONE-TIME · AWARD

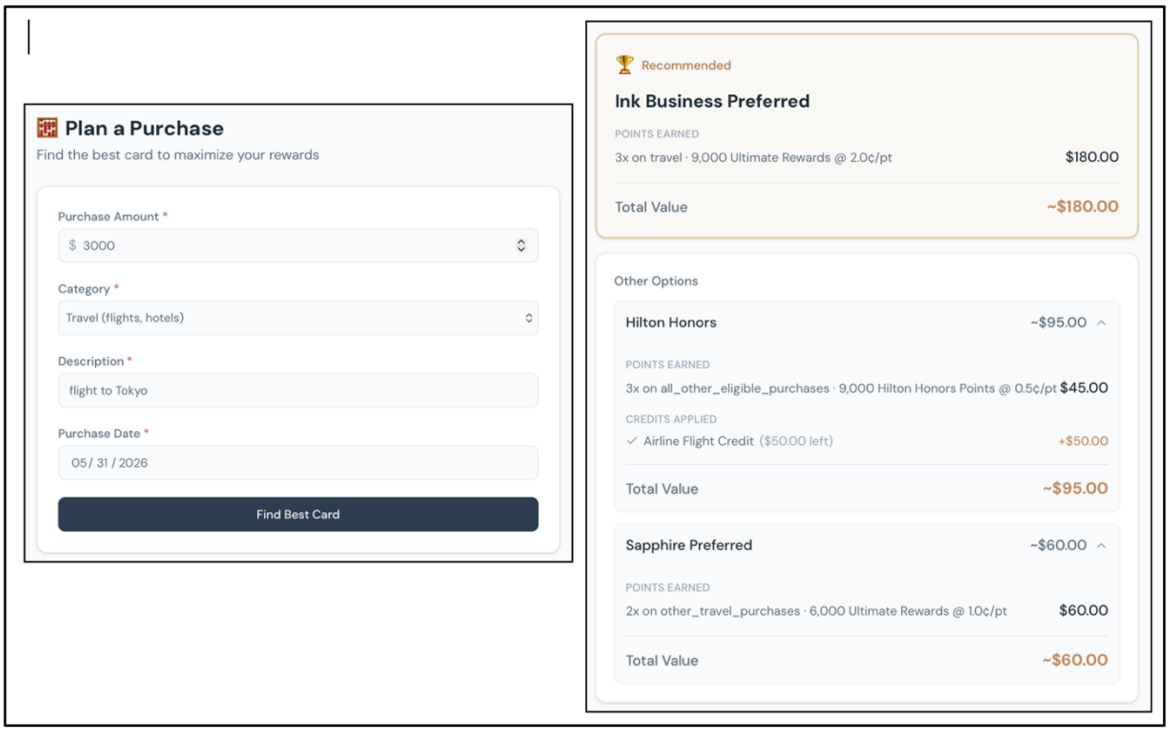
Rewards Ultimate Rewards

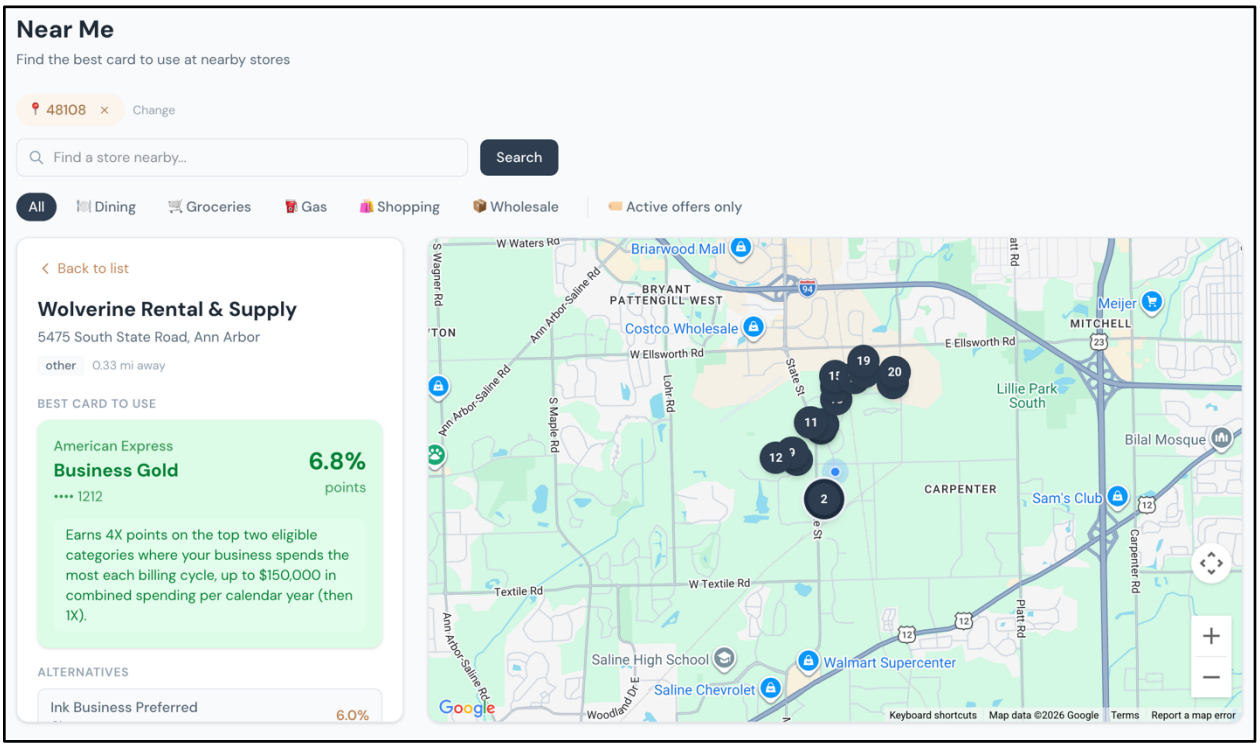
Travel Purchased Through Chase Travel℠	5x	≈ 5% cashback
Dining	3x	≈ 3% cashback
Online Grocery Purchases	3x	≈ 3% cashback
Select Streaming Services	3x	≈ 3% cashback
Other Travel Purchases	2x	≈ 2% cashback
Everything else	1x	≈ 1% cashback

Applies to travel booked through Chase Travel℠. • Includes eligible delivery services and takeout. • Excludes Target, Walmart, and wholesale clubs. • Applies to travel not booked through Chase Travel℠. • Base rate on all other purchases. Ultimate Rewards points can be redeemed for cash at $0.01 per point. Value can be higher for travel redemptions (e.g., 1.25 cents per point through Chase Travel for points earned before June 23, 2025, and redeemed before October 26, 2027) or through transfer partners (up to $0.02 per point).

Figure A4. RAG-Based Card Recommendation Outputs Across Planning Across Multiple Contexts.

The planner and location-aware modules both select the card with the highest estimated return and provide transparent value attribution. Performance is strongest when earning rates and valuations are complete, underscoring the importance of accurate upstream extraction.





Appendix B: Log of AI Interactions

Gate 1 — Document Style Review

Used AI Portfolio Coach (Claude) to complete a seven-item form and style checklist covering column format, font type, font size, title page numbering, Roman numeral pagination, Arabic pagination, and orphaned headers. All seven items confirmed.

Gate 2 — Title Page Review

Worked with AI Portfolio Coach to revise the report title from "AI Credit Card Rewards Assistant" to "Retrieval-Augmented Generation for Personal Finance: An AI-Assisted System for Credit Card Benefit Tracking and Recommendation." Removed program name from Northwestern affiliation. Confirmed professional framing, superscript conventions, and lasting contact information.

Gate 3 — Introduction, Problem Statement, and Abstract

Worked with AI Portfolio Coach to revise the Introduction opening to lead with the problem (cognitive burden, value leakage). Drafted key contribution statement for Paragraph 2. Added results glimpse and roadmap. Revised Abstract against five-beat structure (problem, method, data, results, implications). Generated Figure 1 conceptual overview diagram and keyword list.